

Design and Development of a Manual Tomato Slicer Using Motion Amplification Mechanisms

Mohammad Sharif
MMAE Department
Illinois Institute of Technology
msharif4@hawk.illinoistech.edu

Abstract—This report documents the design, analysis, and iterative development of a manually actuated tomato slicer intended for consumer-scale use. The primary objective was to produce clean, repeatable slices without crushing the tomato while maintaining stability and controlled force application. A review of existing patents showed that relative motion between the blade and food improves slicing quality but increases mechanical complexity. Guided by these findings, multiple linkage-based mechanisms were evaluated through cardboard modeling, CAD analysis, and four physical prototypes. Blade geometries were experimentally tested using controlled loading with scale weights, identifying serrated blades as the most effective cutting profile. The final prototype employs a bell crank mechanism to amplify slicing motion while limiting compressive loading, demonstrating the feasibility of a compact, manually operated slicing device.

I. INTRODUCTION

Tomatoes are mechanically fragile and prone to crushing when subjected to high compressive loads. While industrial tomato slicers often use guided carriages to create a shearing action, most consumer products rely on push-through blade grids, which frequently deform soft produce. The goal of this project was to design a manual tomato slicer that prioritizes amplified relative motion and shear-dominated cutting rather than direct compression.

The functional requirements were:

- Stable base during operation
- Clean slicing without crushing
- Large relative motion between blade and tomato
- Controlled and repeatable manual force input

The design approach emphasized kinematic amplification through mechanical linkages rather than motorized actuation. The final design successfully integrated amplified relative motion into a low-cost commercial tomato slicer (Fig. 1).

II. PRIOR ART AND CONCEPT INSPIRATION

A comprehensive review of prior art was conducted to understand existing approaches to tomato slicing and to identify effective mechanisms for producing clean cuts. Over a dozen patents were analyzed; however, only the most relevant designs that directly informed the final mechanism are discussed here.

Patent US4625607A (Fig. 2) describes a sliding carriage system in which the food is translated relative to a stationary blade array, producing a predominantly shearing cut. This approach directly motivated the decision to prioritize relative motion between the tomato and blade rather than



Fig. 1: Final manual tomato slicer prototype incorporating a bell crank mechanism for motion amplification.

compressive loading. While effective in producing uniform slices, the mechanism is mechanically complex, bulky, and less suitable for consumer-scale products. Additionally, the carriage advances in the same direction as the applied force, whereas reversed displacement relative to input could further improve shearing efficiency and reduce crushing.

A review of existing consumer products revealed a commercially available slicer with strong similarity to this patent architecture, confirming the practical viability of carriage-based slicing while also reinforcing concerns regarding complexity, cleaning difficulty, and manufacturability.

Simpler push-through grid slicers were also examined. These designs rely primarily on compressive forces to drive the tomato through a fixed blade array, which significantly increases the likelihood of crushing soft produce. As a result,

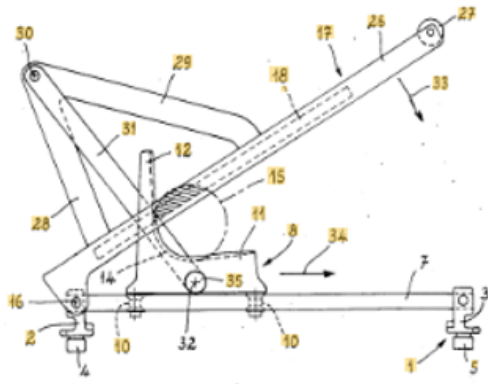


Fig. 2: Sliding carriage slicing concept adapted from US4625607A, illustrating coordinated displacement and cutting motion.

these designs were deemed unsuitable for the intended application.

III. BLADE SELECTION AND EXPERIMENTAL EVALUATION

Blade geometry was identified as a critical factor influencing slicing quality, required force, and the likelihood of crushing. A total of three blade experiments were conducted to systematically evaluate blade profiles and determine an optimal configuration for the slicer.

A. Experiment 1: Qualitative Blade Comparison

An initial qualitative experiment compared a butter knife, a straight-edge sharp knife, and a serrated steel blade across multiple tomato varieties. This test demonstrated that serrated blades consistently produced cleaner cuts with reduced deformation of the tomato interior. Straight-edge blades required higher forces, while blunt blades caused significant crushing. Based on these results, all subsequent testing focused exclusively on serrated blades.

B. Experiment 2: Serration Profile Evaluation

A second experiment evaluated multiple serrated saw blades with varying tooth geometries. A total of five blades were tested: one with small serrations, three with medium serrations, and one with large serrations. The results showed that blades with medium serration profiles performed best overall. Large serrations introduced excessive resistance during cutting, while small serrations were prone to pulp and seed buildup, leading to increased friction and blade slipping.

C. Experiment 3: Controlled Force Testing

Because the initial experiments relied on manual force, a final experiment was conducted using controlled loading to improve repeatability and validity. Scale weights were applied symmetrically to both sides of each blade, ensuring that only horizontal cutting force was introduced (Fig. 3). Each blade was tested three times at 100 g, 200 g, and 400 g.

The results confirmed that blades with medium serration profiles consistently required lower forces and produced



Fig. 3: Scale weight inserts to apply repeatable horizontal cutting forces.

cleaner slices. The two best-performing blades were identified as the BiM S0214 Vermont America blade (Fig. 4) and a black-coated blade, both featuring medium tooth size and moderate teeth-per-inch (TPI). These blades were therefore selected for integration into all subsequent prototype designs.

IV. PROTOTYPE DEVELOPMENT

A. Prototype 1: Crank-Slider Feasibility Model

The first prototype was laser cut and fabricated solely to evaluate the feasibility of a basic crank-slider mechanism (Fig. 5). The objective of this prototype was not performance optimization but verification that rotary input could be reliably converted into linear slicing motion.

B. Prototype 2: Dual Gear, Scotch Yoke, and Scissor Linkage

Prototype 2 and all subsequent prototypes were fabricated using 3D printing to enable greater geometric complexity and rapid iteration. This design introduced a dual gear system with a 5:1 gear ratio, where the smaller gear drove a scotch yoke that actuated a scissor linkage (Fig. 6). The linkage successfully amplified motion by approximately three times relative to the input displacement.

Despite the achieved amplification, this configuration suffered from excessive part count and poor performance when oriented vertically. Gravitational loading caused the linkage to bind, reducing reliability. Adding an additional slider to support the scissor linkage significantly improved motion consistency; however, the added components increased complexity and contradicted the objective of minimizing part count.

C. Prototype 3: Simplified Crank and Reduced Gear

To address the complexity of Prototype 2, Prototype 3 reduced both part count and unused geometry (Fig. 7). The large gear was reduced to approximately one quarter of its original size, as much of the gear circumference was not utilized during operation. The scissor linkage was replaced with a



Fig. 4: BiM S0214 Vermont America blade

simple crank mechanism to improve reliability. Additionally, a dedicated handle was added to the center of the large gear using a keyed interface to ensure positive torque transmission.

While these changes significantly improved robustness and ease of use, the simplified mechanism produced insufficient motion amplification. As a result, the displacement generated was inadequate to drive the slicing operation effectively.

D. Prototype 4: Integrated Housing and Sliding Carrier

Prototype 4 translated the previous mechanisms into a fully integrated 3D assembly featuring a housing, a proper handle, and a sliding carrier designed to hold the tomato as it moved along a rail (Fig. 8). This prototype marked the first attempt to combine the slicing mechanism and food handling system into a single device.

Testing revealed several critical shortcomings. The mechanism was located on only one side of the assembly, resulting in insufficient force to drive the sliding carrier under load. The carrier itself was overly bulky, increasing friction and resistance. Additionally, blades were inserted through a simple slot, which failed to secure them adequately during slicing.



Fig. 5: Prototype 1: Laser-cut crank-slider mechanism used to validate rotary-to-linear motion.

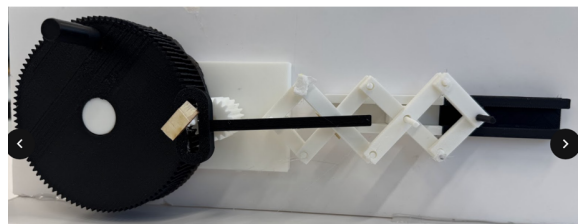


Fig. 6: Prototype 2: 3D-printed dual gear system driving a scotch yoke and scissor linkage.

These issues demonstrated the need for significantly greater motion amplification and improved blade retention in subsequent designs.

V. FINAL PROTOTYPE: BELL CRANK AMPLIFICATION

The final prototype implements a bell crank mechanism to achieve significant motion amplification through differing input and output arm lengths. A short input arm connected to the handle drives a longer output arm, producing increased linear displacement of the slicing carriage with limited user input (Fig. 9).

Several critical design changes were introduced relative to earlier prototypes. A dedicated ergonomic handle attachment was added at the top of the input arm to allow proper torque application and controlled operation. The slicing system was upgraded from a two-blade configuration to a four-blade array to improve slicing consistency and reduce localized loading. Hacksaw blades were selected to allow a pin to pass through both the blade stack and the handle assembly, significantly improving blade retention and alignment during operation.

To address force imbalance observed in earlier designs, the bell crank mechanism was implemented symmetrically on both sides of the device. This dual-sided actuation improved load distribution and enabled reliable translation of the carriage under cutting loads. The tomato carriage was substantially lightweighted, and a domed cutout was introduced to better conform to the tomato profile, improving seating stability while reducing unnecessary mass.

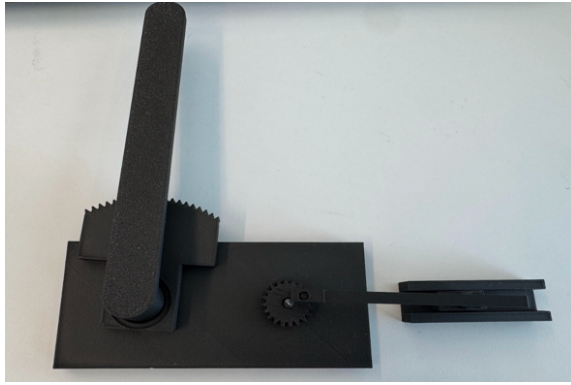


Fig. 7: Prototype 3: Reduced gear and simplified crank mechanism with integrated handle.

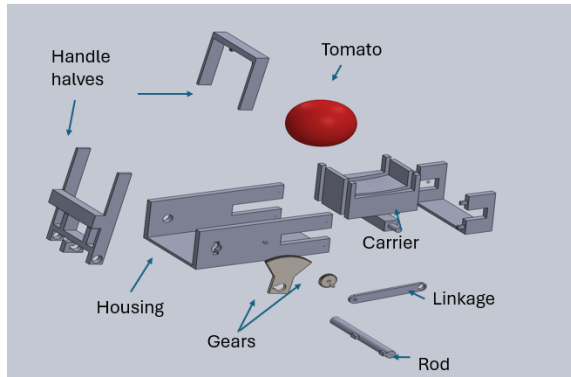


Fig. 8: Prototype 4: CAD of Integrated housing with sliding carrier and single-sided drive mechanism.

These modifications resulted in greatly increased slicing displacement and improved overall performance while maintaining manual operability.

VI. DISCUSSION

The development process demonstrated that effective tomato slicing in a manual device depends on both blade geometry and kinematic design. Serrated blades with medium tooth profiles minimized required cutting force and reduced crushing, while motion amplification mechanisms enabled shear-dominated slicing rather than compression.

Early crank-slider prototypes were useful for validating fundamental motion conversion but failed to generate sufficient displacement. Scissor linkage systems achieved higher amplification but introduced excessive part count, alignment sensitivity, and gravity-induced binding. The bell crank mechanism provided a superior balance between amplification, reliability, and structural simplicity, particularly when implemented symmetrically.

Despite the improvements achieved in the final prototype, several limitations remain. Component retention within the bell crank assembly relies primarily on friction and press fits, which can allow loosening during repeated use. Gear engagement within the housing is also insufficiently constrained, leading to occasional slippage. While motion amplification was

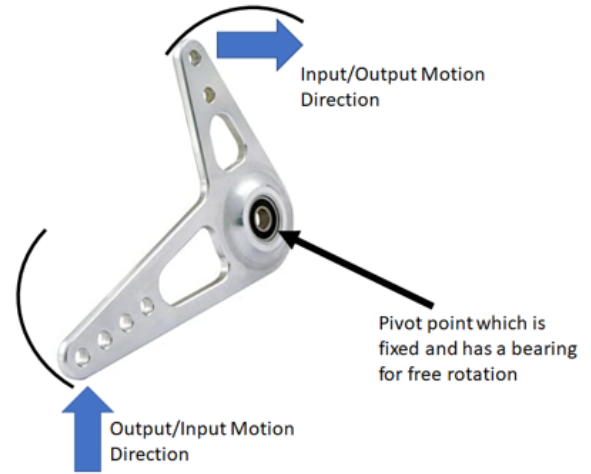


Fig. 9: Bell crank mechanism for amplified slicing motion.

significantly improved, the bell crank assembly still contains more parts than desired for a consumer-oriented design.

VII. CONCLUSIONS

This project demonstrated that clean, repeatable tomato slicing can be achieved in a manual consumer device by prioritizing relative motion, shear-based cutting, and controlled force application. Through systematic prototyping, blade experimentation, and kinematic refinement, a bell crank-based mechanism was developed that provides substantial motion amplification while limiting compressive forces on the tomato.

The final prototype successfully integrates a four-blade slicing array, dual-sided bell crank actuation, and a lightweight, profile-matched carriage. While further refinement is required to reduce part count and improve component retention, the design establishes a strong mechanical foundation for future development toward a robust, manufacturable consumer product.

ACKNOWLEDGMENT

The author acknowledges the use of CAD modeling, rapid prototyping, and experimental testing methods throughout the development of this project.